MIT SCHOOL OF COMPUTING
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Subject: Programming Languages II Roll No.: 44 Class: SOC-9 Batch: B
Name of the Experiment: Assignment-02 Exp. No.:

Performed on:
Submitted on:

| Marks | Teacher's Signature with date |
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| 10 | |

Q1) List the python sequence data type and state one key characteristics of each with respect to mutability or ordering?

Ans) i) List :- Ordered and mutable (elements can be changed)
ii) Tuple :- Ordered and immutable (cannot be modified)
iii) String :- Ordered and immutable sequence of characters
iv) Set :- Unordered, mutable, store unique elements only.
v) Dictionary :- Ordered (Python 3.7+), mutable, stores data as key value pairs

Q2) Explain how indexing and slicing work in python sequences. Illustrate how they differ when applied to string and list?

Ans) Indexing :- It is used to access a single element using its position (start from 0)

Slicing :- It is used to access a part of the sequence using [start : end : step]

Example :-
s = "Python"
I = [10, 20, 30, 40]

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

Indexing:

s[1] → 'y'

I[2] → 30

Slicing:

s[1:4] → 'yth'

I[1:3] → [20, 30]

Difference :-

→ String cannot be modified after slicing (immutable)

→ List can be modified after slicing (mutable)

Q3] Write a python program that:
a) Uses of indexing and slicing on a string list, and
b) Applies atleast two built-in function to apply display meaningfull results

Ans] text = "Rushabh"
print ("first character : ", text [0])
print ("substring : , text [0 : 3]
print ("length of string : ", len (text))
print ("Uppercase : ", text. upper ())

Output:
First character : 15
Substring : Rushabh
length of string : 7
Uppercase : RUSHABH

Q4] Analyse the difference between equality (==) and identity (is) when used with sequence type. Discuss how object mutability and memory reference affect program behaviour.?

Ans] Equality (==) checks whether values are same
Identity (is) checks whether both variable point to the same memory location.

Example :-

| | |
|---|---|
| a = [1, 2, 3] | Output: |
| b = [1, 2, 3] | True |
| point (a == b) | False |
| print (a is b) | |

Explanation :-
- lists are mutable and stored in different memory locations
- (==) compares content
- (is) compares memory reference

Q5) Evaluate the suitability of lists, sets and dictionaries for string and managing unique student records. Justify which data structure is most efficient and why?

Ans) List : Allows duplicates, slower searching.
Set : stores only unique value, no indexing
Dictionaries : stores unique student IDs as keys and detail as value
Most efficient - Dictionary, because it allow fast access, unique keys and structured data storage.

Q6) Design and implement a python program that uses :

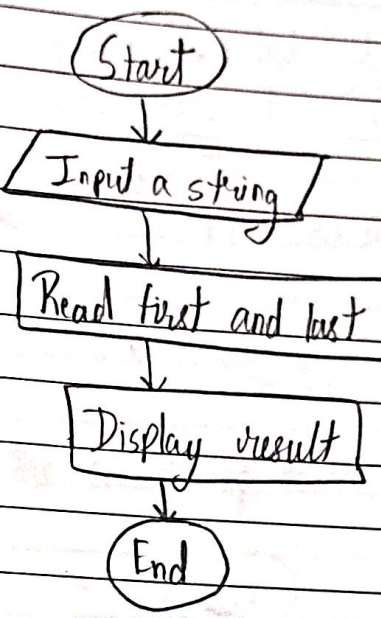
Ans)
```
students = {101: "Amit", 102: "Neha", 103: "Rahul"}
print ("student 101 :", students[101])
print ("Total students :", len (students))
print ("Student IDs :", list (students.key ()))
```

Output :
Student 101 : Amit
Total students : 3
Student IDs : [101, 102, 103]

87) Write a program to print the first and last characters of a string?

Ans s = input ("Enter a string :")
print ("First characters :", s[0])
print ("Last characters :", s[-1])

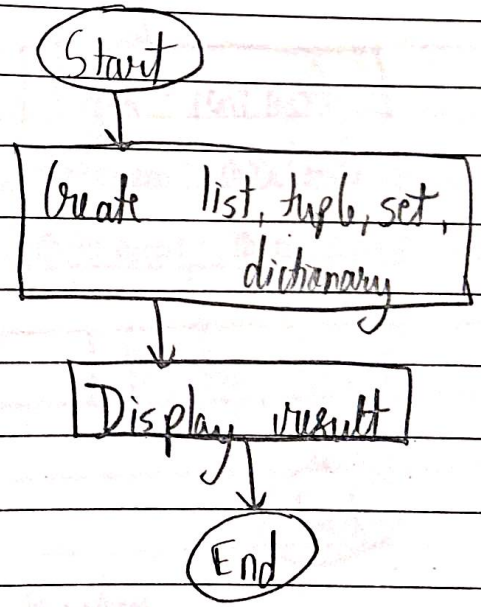
→ Output : First characters : p
Last characters : n




→ Algorithm :-
1) Start
2) Read a string
3) Find the first characters
4) Find the last characters
5) Display result
6) Stop

88) Write a program to create a list, tuple, set and dictionary?

Ans list = [1, 2, 3]
tuple = (4, 5, 6)
string = {7, 8, 9}
dictionary = {1 : "One", 2 : "Two"}
print (list, tuple, string, dictionary)




→ Algorithm :-
1) Start
2) Create a list, tuple, set, dictionary
3) Display result
4) Stop

→ Output :- [1, 2, 3]
(4, 5, 6)

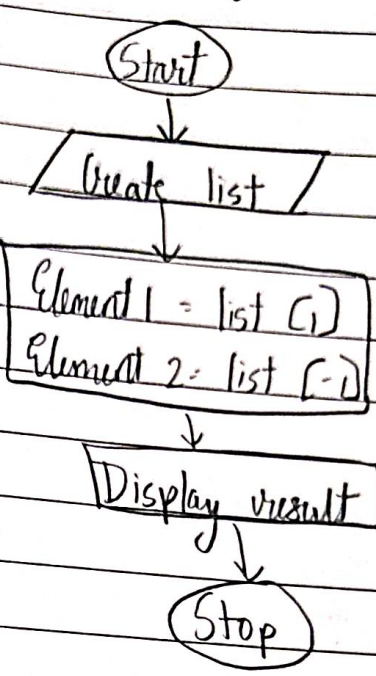
Q9) Write a program to understand the concept of indexing

Ans) array = [10, 20, 30, 40]
print ("Element at index 1:" array [1])
print ("Element at index -1:" array [-1])

→ Output : Element at index 1 : 20
Element at index -1 : 40

→ Algorithm :-

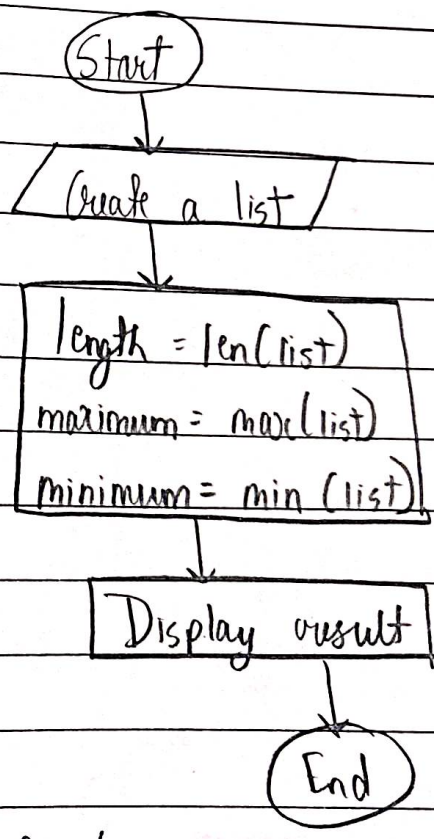
1) Start
2) Create a list of element
3) Access element using positive and negative index
4) Display the result
5) Stop




Q10) Write a program to understant the concept of built in-functions

Ans) num = [5, 2, 9, 1]
print ("length :", len (num))
print ("Maximum :", max (num))
print ("Minimum :", min (num))
print ("Sorted list", sorted (num))

→ Output : length : 4
Maximum : 9
Minimum : 1
Sorted list : [1, 2, 5, 9]

→ Algorithm:

1) Start
2) Grade list
3) Use len() function, max() and min() function
4) End